Informatica

Informatica® PowerExchange for Tableau 10.5.10

User Guide

Informatica PowerExchange for Tableau User Guide
10.5.10
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Table of Contents

Preface 5

Informatica Resources 5

- Informatica Network 5
- Informatica Knowledge Base 5
- Informatica Documentation 5
- Informatica Product Availability Matrices 6
- Informatica Velocity 6
- Informatica Marketplace 6
- Informatica Global Customer Support 6

Chapter 1: Introduction to PowerExchange for Tableau 7

PowerExchange for Tableau Overview 7

Introduction to Tableau 7

PowerExchange for Tableau Implementation 8

PowerExchange for Tableau Example 8

Chapter 2: PowerExchange for Tableau Configuration 9

PowerExchange for Tableau Configuration Overview 9

Prerequisites 9

After You Upgrade 10

Chapter 3: Tableau Connections 11

Tableau Connection Overview 11

Tableau Connection Properties 12

- Content URL 12

Creating a Tableau Connection 13

Rules and Guidelines for Tableau Connections 13

Chapter 4: Tableau Data Objects 14

Tableau Data Objects Overview 14

Tableau Data Extract File 14

Tableau Packaged Workbook File 15

- Workbook Templates 15

Tableau Data Object Properties 20

- Creating a Tableau Data Object 21

Tableau Data Object Write Operation Properties 21

- Creating a Tableau Data Object Write Operation 22

Chapter 5: Tableau Mappings 24

Tableau Mappings Overview 24

Tableau Mapping Example. 24

Appendix A: Data Type Reference. 26

Data Type Reference Overview. 26

Tableau and Transformation Data Types. 26

Decimal Data Type. 28

Duration Data Type. 28

Index. 29

Preface

Use the *Informatica® PowerExchange® for Tableau User Guide* to learn how to read data from and write data to Tableau by using the Developer tool. Learn to create a Tableau connection, develop, and run mappings in the native environment.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Network

The Informatica Network is the gateway to many resources, including the Informatica Knowledge Base and Informatica Global Customer Support. To enter the Informatica Network, visit https://network.informatica.com.

As an Informatica Network member, you have the following options:

- Search the Knowledge Base for product resources.
- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica maintains documentation for many products on the Informatica Knowledge Base in addition to the Documentation Portal. If you cannot find documentation for your product or product version on the Documentation Portal, search the Knowledge Base at https://search.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Product Availability Matrices

Product Availability Matrices (PAMs) indicate the versions of the operating systems, databases, and types of data sources and targets that a product release supports. You can browse the Informatica PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services and based on real-world experiences from hundreds of data management projects. Informatica Velocity represents the collective knowledge of Informatica consultants who work with organizations around the world to plan, develop, deploy, and maintain successful data management solutions.

You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that extend and enhance your Informatica implementations. Leverage any of the hundreds of solutions from Informatica developers and partners on the Marketplace to improve your productivity and speed up time to implementation on your projects. You can find the Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through the Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
https://www.informatica.com/services-and-training/customer-success-services/contact-us.html.

To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

Introduction to PowerExchange for Tableau

This chapter includes the following topics:

- PowerExchange for Tableau Overview, 7
- Introduction to Tableau, 7
- PowerExchange for Tableau Implementation, 8
- PowerExchange for Tableau Example, 8

PowerExchange for Tableau Overview

You can use PowerExchange for Tableau to connect to Tableau from Informatica.

You can integrate and transform data from sources, such as flat files, databases, and applications to generate a Tableau data extract (TDE) file. You can also create a Tableau packaged workbook (TWBX) and publish the generated file to Tableau.

When you connect to sources directly from Tableau, you have to rely on the speed of the underlying data sources. For faster turnaround, offline access, and to share centralized data with multiple users, you can eliminate connecting to data sources directly from Tableau and use the portable TDE file instead.

The TDE and TWBX files are compatible with Tableau products. You can use the TDE or TWBX file in Tableau Desktop to visualize the data extract and identify patterns and trends. You can also use the Tableau connection in a mapping to publish the TDE or TBWX file directly to Tableau Server or Tableau Online.

Note: PowerExchange for Tableau does not support partitioning.

Introduction to Tableau

Tableau software delivers fast analytics, visualization, and rapid-fire business intelligence.

You can use Tableau Desktop to connect to any data, query the data, see patterns, identify trends, and discover visual insights in seconds. You can create interactive visualizations, reports, and dashboards without the need for programming.

Tableau Server is business intelligence that provides browser-based and mobile analytics. You can publish dashboards to Tableau Server, so that other users can interact with the data in a browser or tablet.

Tableau Online is a hosted version of Tableau Server. You can share dashboards with your organization and customers in minutes. The live, interactive views of data in Tableau Online helps you answer your questions in a web browser or tablet.

PowerExchange for Tableau Implementation

To generate a TDE file from the source data, create a Tableau data object and include the data object as a target in a mapping. You can run the mapping or add the mapping to a workflow to process the data, generate the TDE file, and publish the file to Tableau.

When you specify a Tableau workbook template (TWB) for a Tableau target, the Data Integration Service applies the TWB template to the TDE file and generates a Tableau packaged workbook (TWBX) file.

The Data Integration Service integrates with the Tableau data extract API to generate the TDE or TWBX file.

The Data Integration Service uses the Tableau connection to write the TDE or TWBX file to a directory on the machine where the Data Integration Service runs. You can publish the TDE or TWBX file to Tableau Server or Tableau Online. The Tableau Rest APIs publish the TDE or TWBX file to Tableau Server or Tableau Online. When you publish the TDE or TWBX file, the file is available for analysis to multiple users within an organization. You can interact with the data, create reports and dashboards from the data, and visually represent the data.

If you do not want to publish the data to Tableau Server or Tableau Online, you can manually import the TDE or TWBX file from the Data Integration Service machine to Tableau Desktop. You can edit the TDE or TWBX file in Tableau Desktop and later publish the data to Tableau Server or Tableau Online.

PowerExchange for Tableau Example

You are a sales analyst in an enterprise who can access data warehouses or flat files from Tableau Desktop to analyze the data. You want to track the overall growth trend in sales, geographic distribution of sales, and top customers, and present a snapshot of the sales distribution to senior executives.

You can integrate data from multiple sources, filter the data, and make the data available as a TDE file for analysis in Tableau through PowerExchange for Tableau. You can import the TDE file in Tableau Desktop to create interactive, real-time dashboards. The visual representation helps you understand the profitability, with views presented by geography, product category, and customer segment. You can also publish the TDE file to Tableau Server to share a live and interactive dashboard with all the executives in the organization.

CHAPTER 2

PowerExchange for Tableau Configuration

This chapter includes the following topics:

- PowerExchange for Tableau Configuration Overview, 9
- Prerequisites, 9
- After You Upgrade, 10

PowerExchange for Tableau Configuration Overview

PowerExchange for Tableau is installed with Informatica Data Services.

Prerequisites

You must perform the following prerequisites before you can use PowerExchange for Tableau:

- Install and configure the Informatica services.
- Install and configure the Developer tool. You can install the Developer tool when you install Informatica clients.
- Create a Data Integration Service and a Model Repository Service in the Informatica domain.
- Ensure that the version of Red Hat Enterprise Linux is 7 or later for the Data Integration Service that runs on the Linux operating system.

  Note: If the version of Red Hat Enterprise Linux is less than 7, the Tableau connections along with all the other connections might fail.

For more information about product requirements and supported platforms, see the Product Availability Matrix on Informatica Network:
https://network.informatica.com/community/informatica-network/product-availability-matrices

After You Upgrade

After you upgrade Informatica services, perform the following tasks manually:

- Set the Tableau product field in the Tableau connection properties corresponding to the configured setting in the earlier version.
- Set the advance property for a Tableau data object write operation corresponding to the configured setting in the earlier version.

Note: The upgrade steps are not applicable when you upgrade Informatica versions 9.6.1 HotFix 4 and 10.1.

CHAPTER 3

Tableau Connections

This chapter includes the following topics:

- Tableau Connection Overview, 11
- Tableau Connection Properties, 12
- Creating a Tableau Connection, 13
- Rules and Guidelines for Tableau Connections, 13

Tableau Connection Overview

Create a Tableau connection to create a TDE or TWBX file in a directory on the machine where you run the Data Integration Service. You can then use the Tableau connection to publish the TDE or TWBX file to Tableau Server or Tableau Online.

When you create a Tableau connection, you define the connection attributes that the Tableau Rest APIs use to publish the Tableau data extract to Tableau Online or Tableau Server. Enter the connection attributes that are specific to the Tableau product that you want to connect to. You can specify Tableau Desktop, Tableau Server, or Tableau Online in the connection properties.

Specify Tableau Desktop when you want to create a TDE or TWBX file on the local machine. To publish the generated TDE or TWBX file to Tableau Server or Tableau Online, provide the Tableau Server or Tableau Online URL and the user credentials.

Tableau Connection Properties

Use a Tableau connection to connect to Tableau. When you create a Tableau connection, you enter information to access Tableau.

The following table describes the Tableau connection properties:

Property	Description
Name	Name of the Tableau connection.
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Description of the connection. The description cannot exceed 765 characters.
Location	The Informatica domain where you want to create the connection.
Type	Type of connection. Select Tableau.

The following table describes the properties to connect to Tableau:

Connection Property	Description
Tableau Product	The name of the Tableau product to which you want to connect. You can choose one of the following Tableau products to publish the TDE or TWBX file: - Tableau Desktop. Creates a TDE file in the Data Integration Service machine. You can then manually import the TDE file to Tableau Desktop. - Tableau Server. Publishes the generated TDE or TWBX file to Tableau Server. - Tableau Online. Publishes the generated TDE or TWBX file to Tableau Online.
Connection URL	URL of Tableau Server or Tableau Online to which you want to publish the TDE or TWBX file. The URL has the following format: `http://<Host name of Tableau Server or Tableau Online>:<port>`
User Name	User name of the Tableau Server or Tableau Online account.
Password	Password for the Tableau Server or Tableau Online account.
Content URL	The name of the site on Tableau Server or Tableau Online where you want to publish the TDE or TWBX file. Contact the Tableau administrator to provide the site name.

Content URL

You can specify the name of the content URL to point to a specific site on Tableau Server or Tableau Online where you want to publish the TDE file. Specify the site name in the connection properties.

The content URL has the following format: `http://<Host name of Tableau Server or Tableau Online>:<port> /#/site/<Name of the content URL>/View in Tableau Server or Tableau Online>`

For example, if you create a site called *infa* on Tableau Server, the content URL for the site on Tableau Server is: `https://10.50.100.100:6000/#/site/infa/workbooks`

The value you specify for the content URL in the connection properties is *Infa*.

To specify an existing content URL site name on Tableau Server or Tableau Online where you want to publish the Tableau data extract, contact the Tableau Server or Tableau Online administrator.

Creating a Tableau Connection

Before you create a Tableau data object, create a connection in the Developer tool.

1. Click **Window** > **Preferences**.
2. Select **Informatica** > **Connections**.
3. Expand the domain in the **Available Connections**.
4. Select **Enterprise Applications** > **Tableau** and click **Add.**
5. Enter a connection name.
6. Enter an ID for the connection.
7. Optionally, enter a connection description.
8. Select the domain on which you want to create the connection.
9. Select a Tableau connection type.
10. Click **Next**.
11. Configure the connection properties.
12. Click **Test Connection** to verify the connection to Tableau.
13. Click **Finish**.

Rules and Guidelines for Tableau Connections

When you use the command line options to create a connection to Tableau desktop, add the `tabcmdInstallLocation=<location of .tde files>` property in the create connection command to create the connections successfully.

CHAPTER 4

Tableau Data Objects

This chapter includes the following topics:

- Tableau Data Objects Overview, 14
- Tableau Data Extract File, 14
- Tableau Packaged Workbook File, 15
- Tableau Data Object Properties, 20
- Tableau Data Object Write Operation Properties, 21

Tableau Data Objects Overview

A Tableau data object is a physical data object that represents data based on a Tableau resource. The Data Integration Service generates the Tableau data extract file based on the data representation in the Tableau data object.

You can choose to create a Tableau data object or fetch metadata from an existing Tableau data extract file.

The Developer tool adds a default column named tableau_sample when you create the Tableau data object. Based on the columns in the source data, you can open the data object to add the required columns.

When you update a Tableau data extract file, you can either overwrite the file or append data to the existing file. When you append data to an extract file, ensure that the column metadata in the extract file and the Tableau data object are the same.

After you create the Tableau data object, create a Tableau data object write operation and specify the properties for the write operation. You can choose to generate the Tableau data extract file and publish to Tableau Server or Tableau Online. You can also apply a Tableau workbook template to the Tableau data extract file to create a Tableau packaged workbook file. You can create or overwrite data when you publish the Tableau data extract or Tableau packaged workbook file to Tableau.

Tableau Data Extract File

A Tableau Data Extract (TDE) file contains data extracted from an external data source.

The TDE file is a Tableau-specific file format with .tde extension, which contains individual memory-mapped files for each of the columns in the underlying data source. You can use the Tableau connection to publish the TDE file generated in Informatica to Tableau.

Tableau Packaged Workbook File

The Tableau packaged workbook file is a data file with .twbx extension.

Packaged workbooks contain a Tableau workbook along with supporting local file data sources, custom shapes, TDE files, text files, Microsoft Access or Excel files, or background images grouped together in one package. You can publish a packaged workbook from Informatica to Tableau. A workbook can either contain a worksheet or a dashboard.

When you extract data from the source into a TDE file, you can apply a predefined Tableau workbook (TWB) template to the extracted data to create a Tableau packaged workbook (TWBX) in Informatica. The TWB template file is custom made to perform specific analysis on the extracted source data and populates graphs and charts to represent the data. You can publish the TWBX file to Tableau and further modify the generated graphs according to your business needs.

You can apply the following predefined template files for PowerExchange for Tableau to analyse the data extract from Salesforce sources:

- `Salesforce_Opportunity_Analysis.twb`
- `Salesforce_Pipeline_Analysis.twb`
- `Salesforce_Rep_Performance_Analysis.twb`
- `Salesforce_Opportunity_Analysis.twb`
- `Salesforce_Sales_Analysis_by_Region.twb`
- `Salesforce_Sales_Summary.twb`

Note: The current version of PowerExchange for Tableau supports predefined TWB templates for Salesforce data. To use the predefined TWB files and publish the TWBX file, you must use the Salesforce Opportunity object as the source.

Workbook Templates

The templates provide pre-built dashboards for the Salesforce data and use standard Salesforce objects.

You can apply the following template files to a Tableau target to analyze the data extract from Salesforce sources.

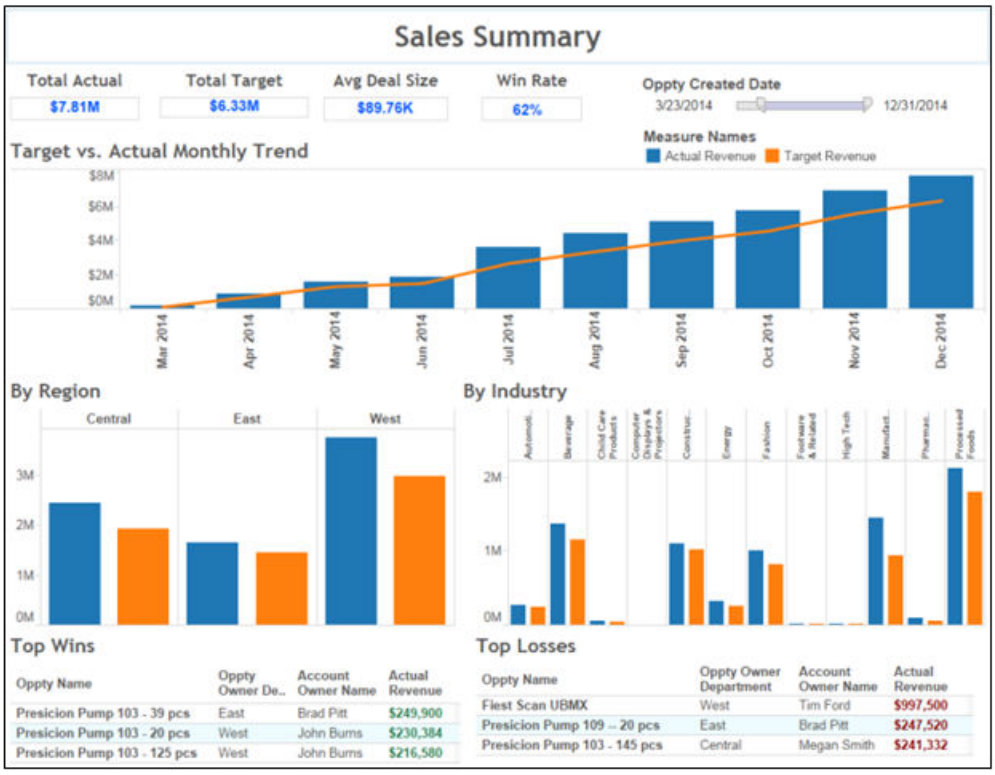
Sales Summary

The sales summary is a dashboard that displays an overview of the sales key performance indicators (KPIs). Use the `Salesforce_Sales_Summary.twb` template to generate the sales summary.

The template uses the following entities to analyze and generate the graph:

- Salesforce Data Objects. Includes Opportunity, Opportunity Owner, Opportunity Creator, Opportunity Account, Account Owner, and Account Creator.
- Measures. Includes Opportunity Expected Revenue and Opportunity Amount.
- Dimensions. Includes Opportunity Owner Department and Account Industry.
- Filter by. Includes Opportunity Create Date.

The following image is a sales summary graph generated from the Salesforce data:



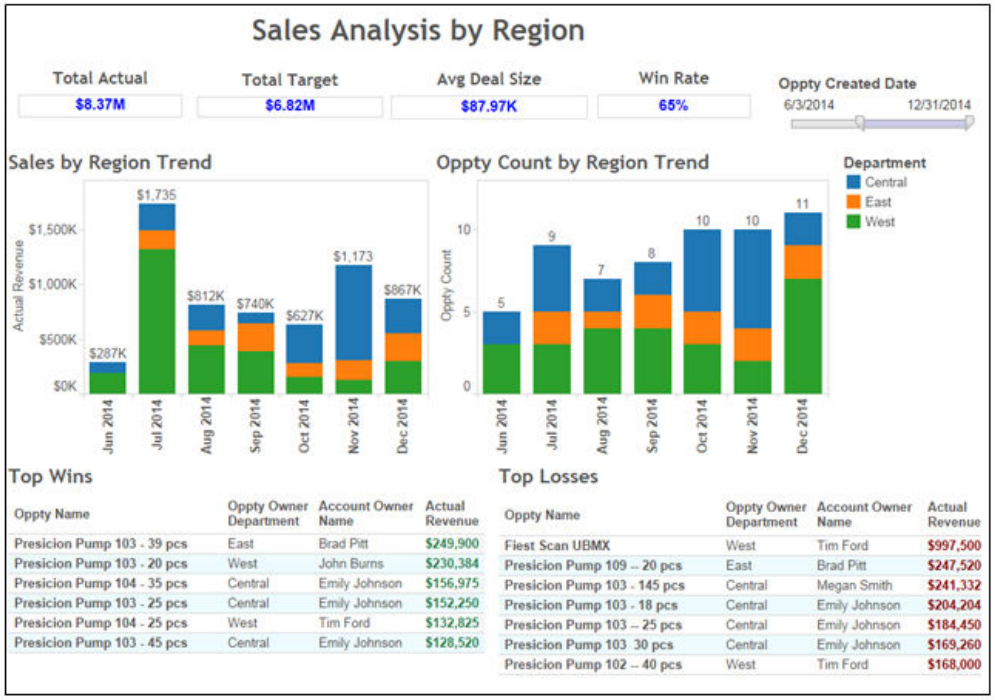
Sales Analysis by Region

Use the `Salesforce_Sales_Analysis_by_Region.twb` template to analyze and display the sales KPIs by region.

The template uses the following entities to analyze and generate the graph:

- Salesforce Data Objects. Includes Opportunity, Opportunity Owner, Opportunity Creator, Opportunity Account, Account Owner, and Account Creator.
- Measures. Includes Opportunity Expected Revenue, Opportunity Amount, and Opportunity Count.
- Dimensions. Includes Opportunity Owner Department.
- Filter by. Includes Opportunity Create Date.

The following image is a sales analysis summary graph generated from the Salesforce data:



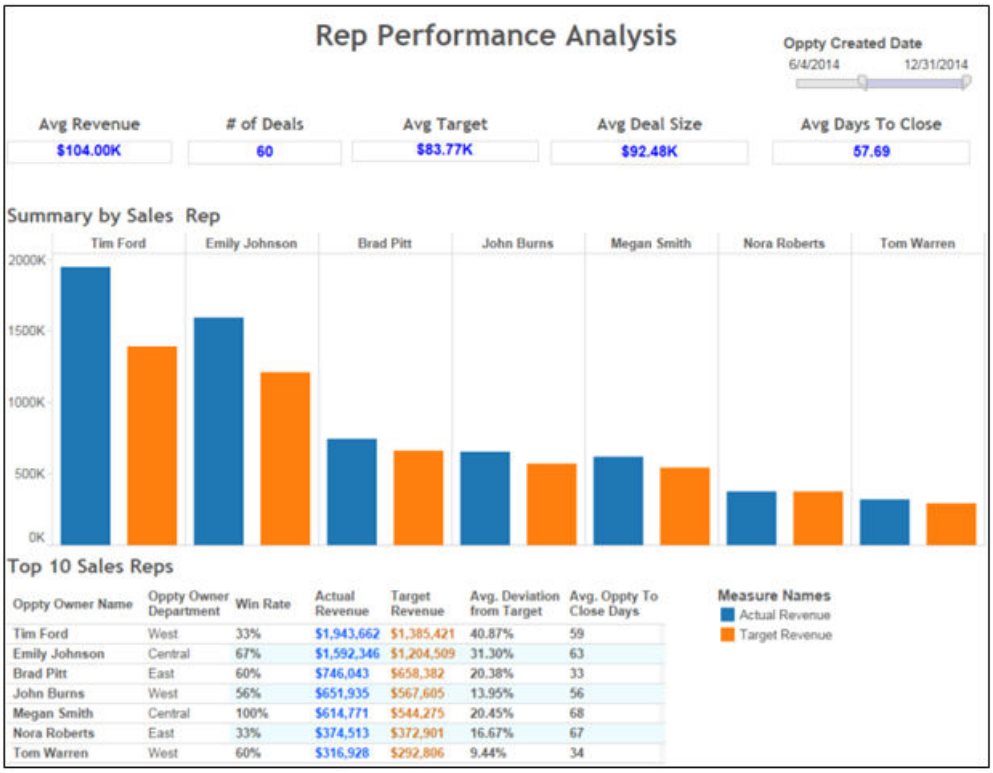
Representative Performance Analysis

Use the `Salesforce_Rep_Performance_Analysis.twb` template to track the performance of the sales team.

The template uses the following entities to analyze and generate the graph:

- Salesforce Data Objects. Includes Opportunity, Opportunity Owner, Opportunity Creator, Opportunity Account, Account Owner, and Account Creator.
- Measures. Includes Opportunity Expected Revenue and Opportunity Amount.
- Dimensions. Includes Opportunity Owner Name.
- Filter by. Includes Opportunity Create Date.

The following image is a sales summary graph generated by region from the Salesforce data:



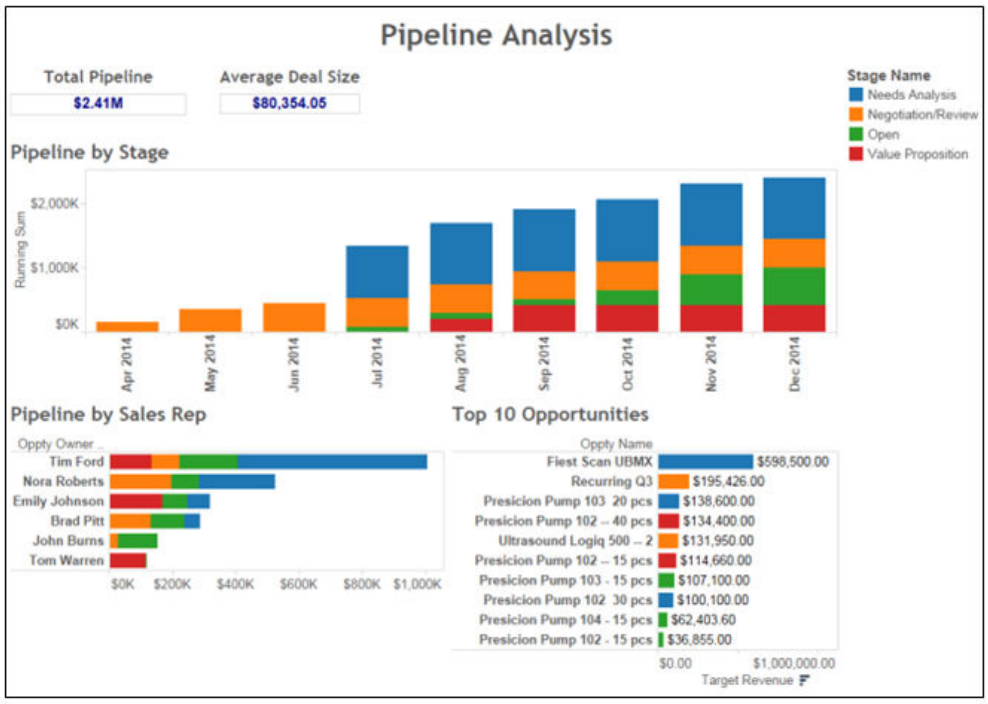
Pipeline Analysis

Use the `Salesforce_Pipeline_Analysis.twb` template to analyze the sales pipeline.

The template uses the following entities to analyze and generate the graph:

- Salesforce Data Objects. Includes Opportunity, Opportunity Owner, Opportunity Creator, Opportunity Account, Account Owner, and Account Creator.
- Measures. Includes Opportunity Expected Revenue.
- Dimensions. Includes Opportunity Owner Nam and Opportunity Stage Name.
- Filter by. Includes Opportunity Create Date.

The following image is a pipeline analysis graph generated from the Salesforce data:



Opportunity Summary

Use the `Salesforce_Opportunity_Analysis.twb` template to understand from where you get the sales opportunities.

The template uses the following entities to analyze and generate the graph:

- Salesforce Data Objects. Includes Opportunity, Opportunity Owner, Opportunity Creator, Opportunity Account, Account Owner, and Account Creator.
- Measures. Includes Opportunity Expected Revenue.
- Dimensions. Includes Opportunity Type and Account Industry..
- Filter by. Includes Opportunity Create Date and Account Billing Address Country.

The following image is a sales opportunity summary generated from the Salesforce data:

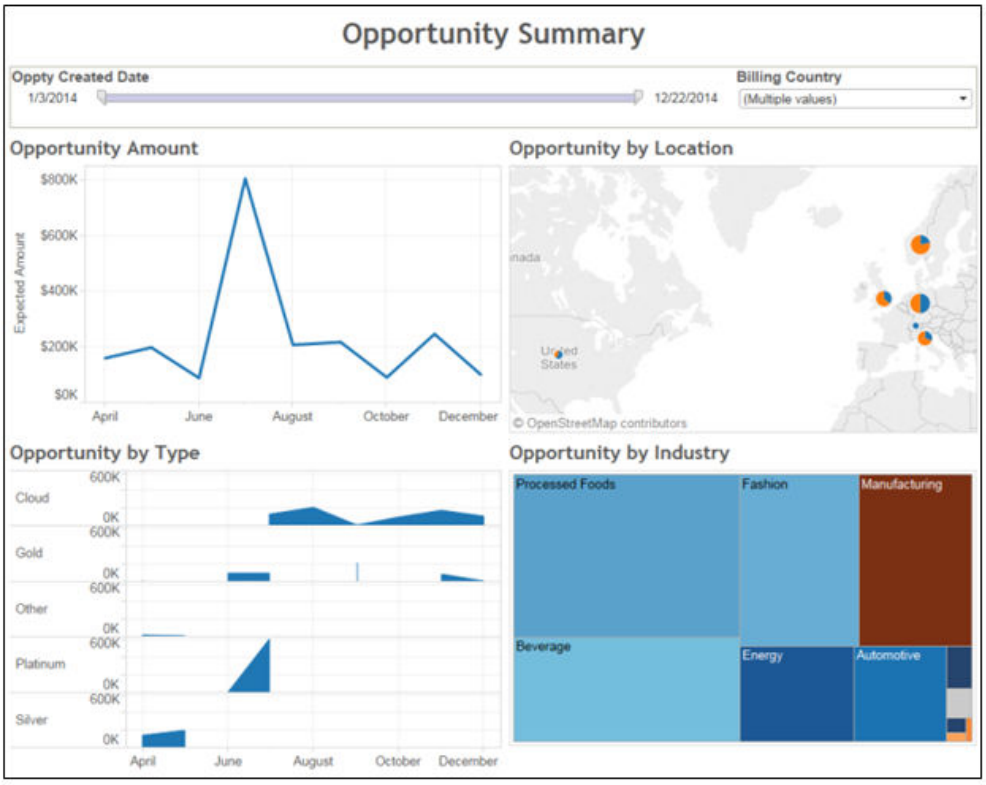


Tableau Data Object Properties

Specify the data object properties when you create a Tableau data object.

The following table describes the properties that you configure for a Tableau data object:

Property	Description
Name	Name of the Tableau data object.
Location	The project or folder in the Model Repository Service where you want to store the Tableau data object.
Access Type	You can choose one of the following options: - Create an empty metadata object. Creates a Tableau data object with a sample port. - Fetch metadata from file. Imports metadata from the specified Tableau data extract file.
Resource Location	Location of the Tableau data extract file. Required when you specify the access method as File.

Creating a Tableau Data Object

Create a Tableau data object to specify a Tableau resource.

1. Select a project or folder in the Object Explorer view.
2. Click **File** > **New** > **Data Object**.
3. Select **Tableau Data Object** and click **Next**.

   The **New Tableau Data Object** dialog box appears.
4. Enter a name for the data object.
5. Click **Browse** next to the **Location** option and select the target project or folder.
6. Specify the access method. Choose to create a Tableau data object or fetch data from an existing Tableau data extract file.
   - To create a Tableau data object, select **Create an Empty Metadata Object** as the access method.
   - To fetch data from an existing Tableau data extract file, select **Fetch Metadata from File** as the access method.
7. To fetch data from an existing Tableau data extract file, perform the following tasks:
   a. Click **Browse** next to the **Resource Location** option. The Windows **Open** dialog box appears.
   b. Navigate to the location of the Tableau data extract file with .tde extension and click **Open**.

      Ensure that the Tableau data extract file was generated by using the Tableau Data Extract API and contains the Extract object.
   c. Optionally, select the table in the extract file and click **View** to review the column details.
8. Click **Finish**.

   The data object appears under Data Object in the project or folder in the Object Explorer view.

Tableau Data Object Write Operation Properties

Tableau data object write operation properties include run-time and advanced properties that apply to the Tableau data object.

The run-time properties display the name of the Tableau connection. The Developer tool displays advanced properties for the Tableau data object operation in the Advanced view.

The following table describes the advanced properties that you can configure for a Tableau data object write operation:

Property	Description
Target Operation	Creates, appends, or overwrites the TDE file on the local machine, Tableau Server, or Tableau Online, as specified in the Tableau connection properties. Select one of the following options to publish the TDE file: - create. Creates a TDE file. Ensure that a TDE file with the same name does not exist. - append. Adds data to an existing TDE file. - overwrite. Deletes the existing TDE file and creates a new TDE file. The append operation works only if there is an existing TDE file.
Extract File Path	The file path where you want to save the generated Tableau data extract file. Ensure that the file path is on the machine where the Data Integration Service runs. The path is a temporary location for Tableau Server and Tableau Online. Default is the location of the target file specified in the Tableau connection properties.
Extract File Name	Name for the Tableau data extract file with the .tde extension. Default is the file name you specified when creating the target object. If the operation is for Tableau Server or Tableau Online, the file gets deleted after the Data Integration Service publishes the TDE file to Tableau Server or Tableau Online. To publish a TWDX file to Tableau, you must provide the extract file name `salesforce_opportunity.tde` defined in the template.
Project Name	Name of the project within a specific site on Tableau Server or Tableau Online where you want to publish the Tableau data extract. By default, the Tableau connection publishes the TDE file to the default project on the site that you specify on Tableau Server.
Data Source	Name of the Tableau data extract that you want to publish to Tableau Server or Tableau Online. If you do not specify a data source name, the default Tableau data extract file name remains the source name. If you do not specify a data source name, the extract file name remains the data source name. If you do not specify both the data source name or the extract file name, the metadata file name that you specify when creating the Tableau data object remains the data source name.
Workbook Template File Name	Name of the predefined Tableau workbook template (TWB) file name that you want to apply to the TDE file to generate a Tableau packaged workbook file (TWBX). You must provide the TWB name to publish the TWBX file.
Workbook Name	Name for the workbook that you want to publish to Tableau. If you do not specify a workbook name, the name of the TWB template file remains the workbook name.

Creating a Tableau Data Object Write Operation

Create a Tableau data object write operation from a Tableau data object.

Before you create a data object operation, you must create the data object with the resource.

1. Select the data object in the Object Explorer view.
2. Right-click and select **New** > **Data Object Operation**.

   The **Data Object Operation** dialog box appears.
3. Enter a name for the data object operation.
4. Select the type of data object operation. Select **extracts Write** to create a write operation.
5. Click **Add**.

The **Select a resource** dialog box appears.

6. Select the resource for which you want to create the data object operation and click **OK**.
7. Click **Finish**.

   The Developer tool creates the data object operation for the selected data object.

Open the Tableau data object operation to edit the input properties in the Advanced view.

CHAPTER 5

Tableau Mappings

This chapter includes the following topics:

- Tableau Mappings Overview, 24
- Tableau Mapping Example, 24

Tableau Mappings Overview

After you create a Tableau data object operation, you can develop a mapping.

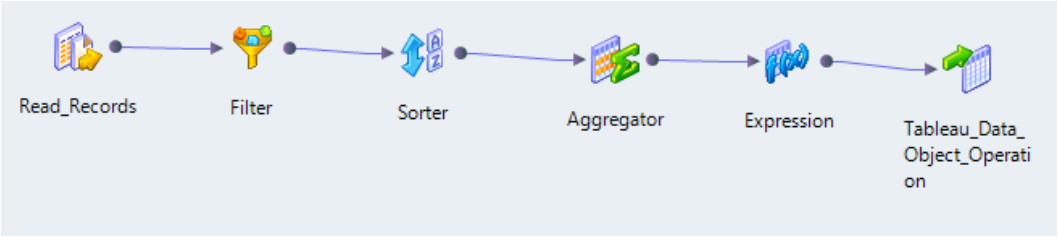
You can add a Tableau data object operation to a Tableau mapping as a target. Validate and run the mapping to integrate and transform source data and generate the Tableau data extract file. You can edit the Tableau data object operation run-time properties to modify the default values. You can also add advanced run-time properties to publish the Tableau data extract file to Tableau Online or Tableau Server.

Tableau Mapping Example

You work in the retail industry, and business analysts in your enterprise need to analyze product sales trends based on region.

The sales record files contain columns with information about products that are sold in multiple outlets and regions. You consolidate the data in the sales record files that you receive through the day. You can then perform transformations based on your requirements.

The following image shows a Tableau mapping:



You use the following objects in the Tableau mapping:

Flat file data object

The source for the mapping is a flat file data object that contains the product sales data.

Create a flat file data object and specify the sales record as the resource for the data object. Source columns in the flat file data object include Region ID, Product ID, Quantity, and Cost. Configure the read properties of the data object.

Transformations

Add transformations to get aggregate data about the product sales in a particular region.

- The Filter transformation filters the data in the sales record files based on the value you specify for the region ID column.
  The Data Integration Service returns the rows that meet the filter condition.
- The Sorter transformation sorts the data in ascending order based on the region ID.
- The Aggregator transformation collects statistics about product sales for a particular region.
  Use the result of the Sorter transformation as an input to the Aggregator transformation. You can increase Aggregator transformation performance with the sorted input option.

Tableau data object write operation

The target of the mapping is a Tableau data object write operation. Create a Tableau data object and then create a Tableau data object write operation to generate the Tableau data extract file.

Specify the absolute path and name of the Tableau data extract file to which you want to write the data. You must then specify the content URL for a specific site on Tableau Server where you want to publish the Tableau data extract file.

When you run the mapping, the Data Integration Service writes the sales information to a target TDE file and publishes the TDE file to Tableau Server. You can then visualize the sales data categorized by region in Tableau Server.

APPENDIX A

Data Type Reference

This appendix includes the following topics:

- Data Type Reference Overview, 26
- Tableau and Transformation Data Types, 26
- Decimal Data Type, 28
- Duration Data Type, 28

Data Type Reference Overview

Informatica Developer uses the following data types in Tableau mappings:

- Tableau native data types. Tableau data types appear in the physical data object column properties.
- Transformation data types. Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Data Integration Service uses to move data across platforms. Transformation data types appear in all transformations in a mapping.

When the reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the Data Integration Service writes to a target, it converts the transformation data types to the comparable native data types.

Tableau and Transformation Data Types

The following table lists the Tableau data types that the Data Integration Service supports and the corresponding transformation data types:

Tableau Data Type	Transformation Data Type	Range and Description
Integer	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Double	Double	Double-precision floating-point numeric value. Precision 15

Tableau Data Type	Transformation Data Type	Range and Description
Date	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision of 29, scale of 9 (precision to nanosecond)
DateTime	Date/Time	Jan 1, 0001 A.D. to Dec 31, 9999 A.D. Precision of 29, scale of 9 (precision to nanosecond)
unicode_string	String, Text, Bigint, or Decimal	The Data Integration Service performs an implicit conversion of String, Text, Bigint, or Decimal to unicode_string: String: - 1 to 104,857,600 characters - Fixed-length or varying-length string Text: - 1 to 104,857,600 characters - Fixed-length or varying-length string Bigint: - 9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 - Precision of 19, scale of 0 - Integer value Decimal: - Precision 1 to 28 digits, scale 0 to 28 - Decimal value with declared precision and scale. Scale must be less than or equal to precision.
char_string	String, Text, Bigint, or Decimal	The Data Integration Service performs an implicit conversion of String, Text, Bigint, or Decimal to char_string: String: - 1 to 104,857,600 characters - Fixed-length or varying-length string Text: - 1 to 104,857,600 characters - Fixed-length or varying-length string Bigint: - 9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 - Precision of 19, scale of 0 - Integer value Decimal: - Precision 1 to 28 digits, scale 0 to 28 - Decimal value with declared precision and scale. Scale must be less than or equal to precision.
boolean	String	1 to 104,857,600 characters. Fixed-length or varying-length string. Valid values are True and False.
duration	String	Valid values for hours are integer values between 0 and 23. Valid values for minutes and seconds are integer values between 0 and 59. If there is no value for any field, specify 0.

Decimal Data Type

When you read data as Decimal in the reader object, use String or Double instead of the Decimal data type for better performance. As Tableau does not support the Decimal data type, you must change the decimal data to string data type that Tableau supports. Change the decimal data type to string in the input wizard of the Tableau data object and char_string or unicode_string in the output wizard of the Tableau data object. To write the decimal data to double data type supported by Tableau, change the decimal data type to double in the input wizard of the Tableau data object. The Data Integration Service performs an implicit conversion of decimal to a comparable native data type, unicode or char_string, that Tableau supports.

Duration Data Type

Duration is specified in days, hours, minutes, seconds, and milliseconds. All the values must be integers. You must change the string data type that arrives from different source fields to a single string value and then map this string value to the duration data type in the target operation.

For example, the Data Integration Service reads data from five different source fields of string data type, such as, 5 days, 10 hours, 21 minutes, and 35 seconds. Use the Expression transformation to concatenate the input string values to a single string value of comma-separated values, such as 5,10,21,35,0. Map the string output received from the Expression transformation to duration data type. Use the single string value as the input value and duration as the output value in the target operation of the mapping.

INDEX

C

connection properties
  content URL 12
content URL
  connection properties 12
creating
  Tableau data object 21
  Tableau data object operation 22

D

data type reference
  overview 26

O

overview
  Tableau data objects 14

P

PowerExchange for Tableau
  configuration overview 9
  data types 26
  overview 7

R

run-time properties
  Tableau data object operation 21

T

Tableau data object
  creating 21
  general properties 20
  overview 14
Tableau data object operation
  creating 22
  properties 21
Tableau mapping
  example 24
  overview 24

U

upgrade
  Informatica services 10

W

workbook templates
  opportunity summary 15
  pipeline analysis 15
  rep performance analysis 15
  sales analysis by region 15
  sales summary 15